

Wachirawit Piyaprapapan

Data & AI Engineer | ML Systems, Data Platforms, Applied AI

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Education

Chulalongkorn University, B.Eng. in Electrical Engineering – Bangkok, Thailand

Aug 2022 – May 2026

- GPAX: 3.44 (Second-class honours)
- Coursework: Data Science, Data Engineering, Estimation, Statistical Learning, Optimization
- Capstone: Generative Video-Based Sky Image Forecasting for Thai Sky Images (SkyGPT, VQ-VAE)

Skills

Languages: Python, JavaScript, SQL, Bash

Software & Frameworks: PyTorch, FastAPI, Next.js, React, Scikit-learn, PySpark, Pandas, SQLModel, spaCy, OpenCV, Selenium

Tools: Docker, Apache Airflow, dbt, PostgreSQL, Grafana, Git

Services & Platforms: GCP (Google Cloud Platform), Vercel, Hugging Face

Experience

AI Engineer Part Time, Hobbit Technologies – Bangkok, Thailand

Feb 2026 – present

- Developed a RAG pipeline using LiteLLM and FastAPI, implementing Pydantic-validated structured outputs that reduced downstream parsing errors.
- Architected a modular Next.js/PostgreSQL platform to generate/evaluate CEFR-aligned content, utilizing NLP for automated grammar correction.
- Scaled LLM services via REST APIs, improving generation consistency through Zero-shot Templating and Chain-of-Thought .

AI Engineer Intern, Hobbit Technologies – Bangkok, Thailand

June 2025 – Aug 2025

- Engineered an end-to-end Computer Vision data engine using OpenCV and Docker, integrating automated data drift detection to trigger active learning loops. This self-training pipeline reduced labeling costs by ~20k THB/year
- Deployed an end-to-end observability stack (Grafana/Prometheus) to monitor Edge AI inference across on-premise IoT sensors, preventing system failure in the production line.

Electrical Engineering Intern, AGC Flat Glass – Bangkok, Thailand

June 2024 – Aug 2024

- Built a data-driven control logic prototype integrating production data into PLC workflows, improving operational efficiency by ~10%.

Projects

Generative Video-Based Sky Image Forecasting for Thai Sky Images

- Implemented and optimized VQ-VAE architecture for atmospheric forecasting by implementing Charbonnier and Perceptual loss functions, EMA codebook updates, and sub-pixel interpolation decoding.
- Achieved a peak validation PSNR of 37.32 dB (+16% over baseline) and compressed channel-sequence representation to ensure stable temporal prediction and reduced computational complexity.

On-Demand Delivery Data Platform & Decision Intelligence System

- Engineered ETL/ELT workflows in Python, PostgreSQL, Airflow, dbt, and Docker to ingest raw JSON events into a structured feature store.
- Added automated schema validation and time-aware model validation to support production-ready ETA and delay prediction.

End-To-End Football Player Value Forecasting & Similarity Recommendation System

- Developed time-series regression and clustering pipelines to forecast player market value and group players by role similarity.
- Turned model outputs into recruitment and benchmarking insights using forecasting error analysis and meaningful cluster structure.